

BM4 Implicit Method

Physical state and one reduced Hairer projection per complete cycle

1 Twelve-stage BM4 base map

Let χ_h be a first-order map and $\chi_h^* = \chi_{-h}^{-1}$ its adjoint. One complete step uses the palindromic composition

$$\Psi_h = \chi_{b_{12}h} \circ \chi_{b_{11}h}^* \circ \cdots \circ \chi_{b_2h} \circ \chi_{b_1h}^*,$$

with alternating adjoint and direct maps in chronological order and

$$(b_1, \dots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

The half-cycle coefficients sum to $1/2$. Palindromy makes the composition self-adjoint and the order conditions give designed global order four. The negative coefficient a_4 is an intentional backward subflow.

2 Physical state and internal workspace

The only accepted state is the physical guiding-centre state $z \in \mathbb{R}^{2N}$. Define

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \frac{1}{2} \begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T.$$

The doubled vector in $\mathbb{R}^{4N}$ is temporary algebraic workspace required by the direct-adjoint splitting and Hairer projection. It is neither an accepted state nor a public state-extension option. Time remains an external integration parameter; no time-conjugate momentum is added to the state.

3 One Hairer projection around the complete cycle

For every complete BM4 step, the method displaces the diagonal embedding by a single multiplier $\mu \in \mathbb{R}^{2N}$,

$$\hat{Y}_n = Ez_n + N\mu, \quad M(\mu) = \Psi_{h,t_n}(\hat{Y}_n), \quad (1)$$

$$r(\mu) = GM(\mu) + 2\mu, \quad z_{n+1} = P(M(\mu) + N\mu). \quad (2)$$

The accepted multiplier is the root $r(\mu) = 0$. Thus the projection is applied once around the full twelve-stage composition, never between its internal stages.

With $J_{\text{BM4}} = D\Psi_{h,t_n}(\hat{Y}_n)$, Newton uses

$$D_\mu r = GJ_{\text{BM4}}N + 2I.$$

The implementation forms J_{BM4} as the exact ordered product of the twelve guiding-centre stage Jacobians, with a centred-difference fallback. Good Broyden is retained as a nonlinear-solver choice; it changes the finite iteration path but not the defining projected map.

4 Single public method

The model exposes only `BM4Implicit`. Its numerical choices are solver controls rather than distinct model classes:

Fixed architectural choice	Value
Projection placement	once around each complete BM4 cycle
Projection formulation	reduced Hairer multiplier
Accepted state	physical $z \in \mathbb{R}^{2N}$
Temporary base-map workspace	doubled state in $\mathbb{R}^{4N}$
Public state extension	none

The configurable controls are the coupling frequency, Newton tolerances, maximum iterations, analytic or finite-difference Newton Jacobian, Newton or Broyden nonlinear solver, progress reporting, and an optional step observer.

5 Fixed-grid lifecycle and diagnostics

The accepted physical state is advanced on the output-independent grid. An off-grid requested sample uses a shadow step from the preceding main node and is discarded after sampling, so output density does not alter the accepted trajectory. Only main steps contribute observer records and diagnostics.

Each accepted step reports nonlinear iterations, residual evaluations, final residual and tolerance, the projection-multiplier norm, coupling frequency, and Jacobian strategy. An optional `ImplicitBM4IntegrationStep` contains the physical input/output, multiplier, solver metrics, and twelve reconstructed base-stage snapshots. Generalized energy may be reconstructed as a diagnostic from those snapshots without introducing a public extended state.

6 Implementation correspondence

The public implementation is `src/simulation/methods/bm4/implicit.py`. The private `_core.py` module owns only the coefficient schedule and twelve-stage base-map traversal. The detailed reduced-projection derivation is retained in `implicit-reduced.tex`; runtime architecture is under `../simulation/`.